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(54) **QUANTUM LIDAR SYSTEM**

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G01S 17/89	(2020.01)

(52) **U.S. Cl.**
CPC ***G01S 17/89*** (2013.01); ***G01S 7/48*** (2013.01)

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CPC G01S 17/89; G01S 7/48
See application file for complete search history.

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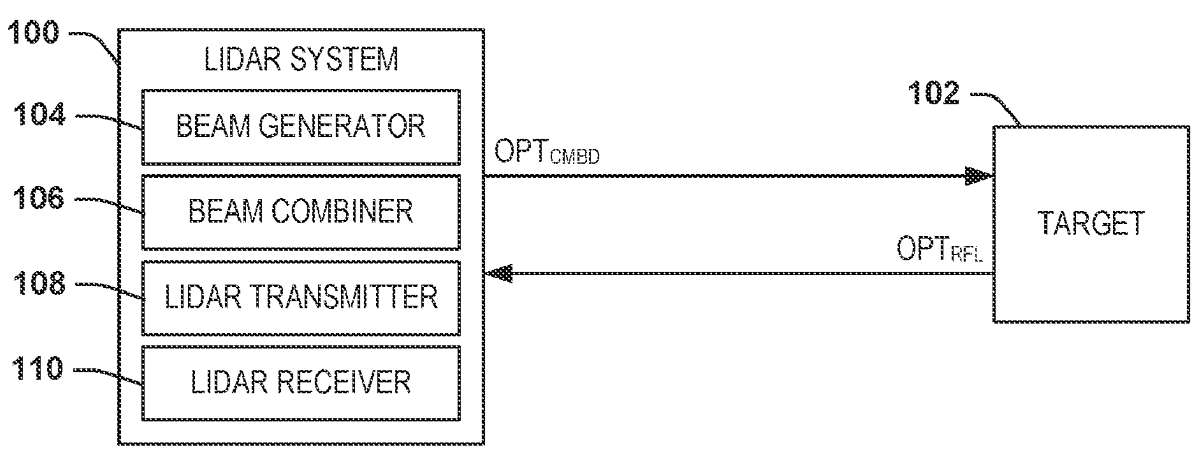
(57) **ABSTRACT**

One example includes a quantum lidar system. The system includes a beam generator configured to generate a signal beam and an idler beam and a beam combiner configured to generate a combined optical beam comprising the signal beam and the idler beam. The system also includes a lidar transmitter configured to transmit the combined optical beam to a target and a lidar receiver configured to receive the combined optical beam and a reflected beam of the combined optical beam reflected from the target to generate lidar data associated with the target.

20 Claims, 3 Drawing Sheets



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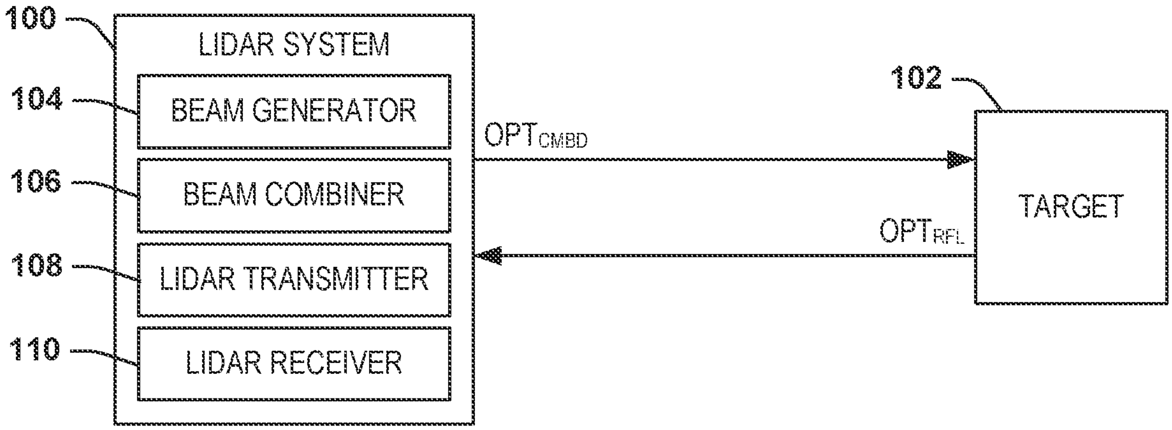
100
LIDAR SYSTEM
104
BEAM GENERATOR
106
BEAM COMBINER
108
LIDAR TRANSMITTER
110
LIDAR RECEIVER
102
OPT_{CMBD}
TARGET
OPT_{RFL}

FIG. 1

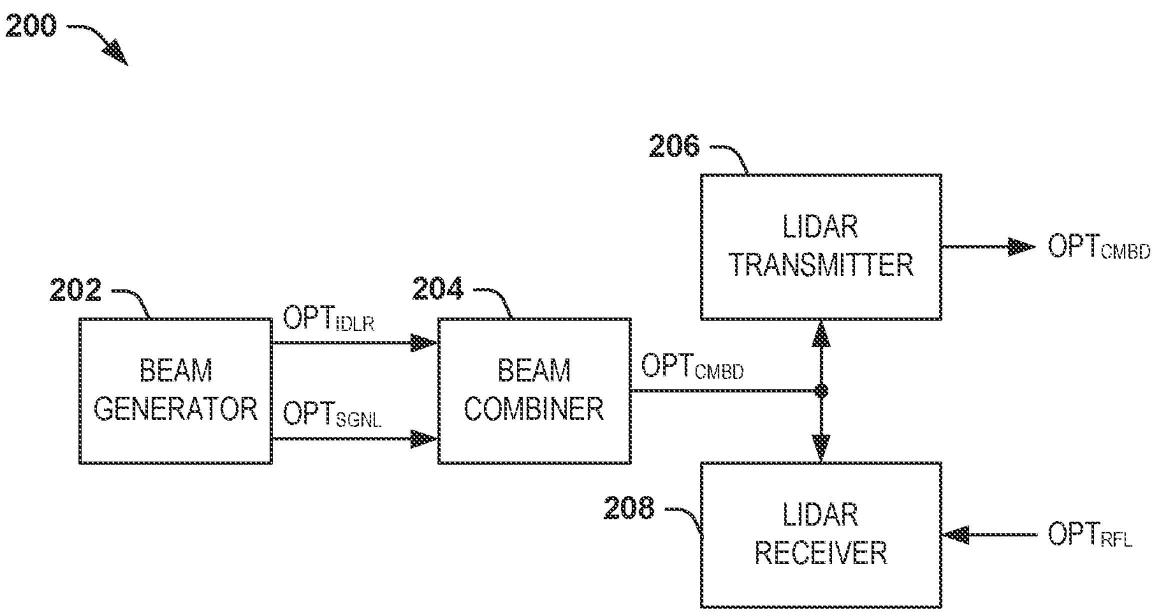
200
206
LIDAR TRANSMITTER
OPT_{CMBD}
202
204
OPT_{IDLR}
BEAM GENERATOR
OPT_{SGNL}
BEAM COMBINER
OPT_{CMBD}
208
LIDAR RECEIVER
OPT_{RFL}

FIG. 2

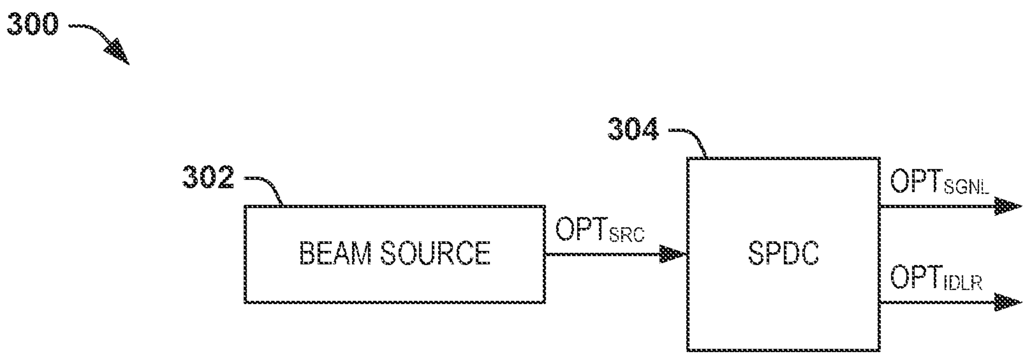
300
302
BEAM SOURCE
OPTSRC
304
SPDC
OPTSGNL
OPTIDLR

FIG. 3

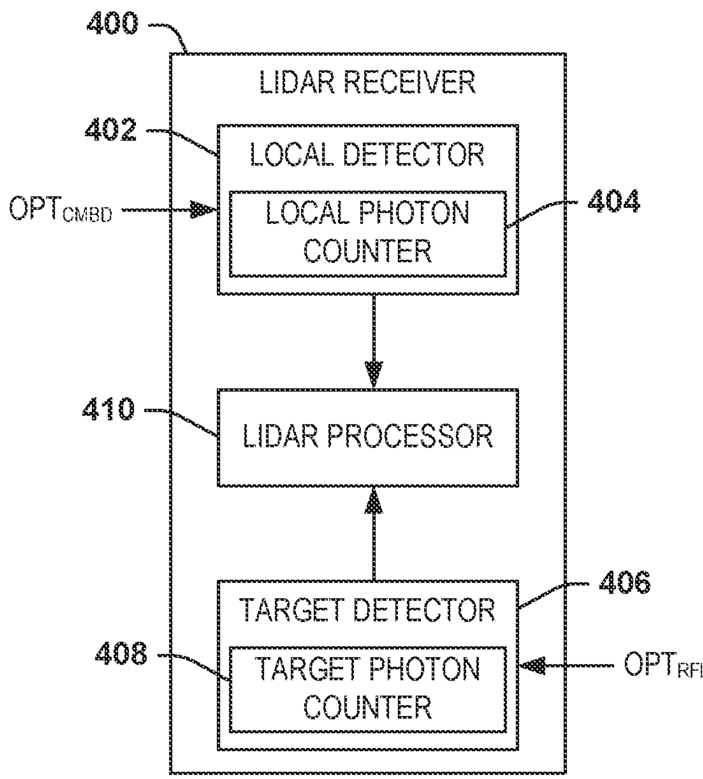
400
LIDAR RECEIVER
402
LOCAL DETECTOR
OPTCMBD
LOCAL PHOTON COUNTER
404
410
LIDAR PROCESSOR
406
TARGET DETECTOR
408
TARGET PHOTON COUNTER
OPTRFL

FIG. 4

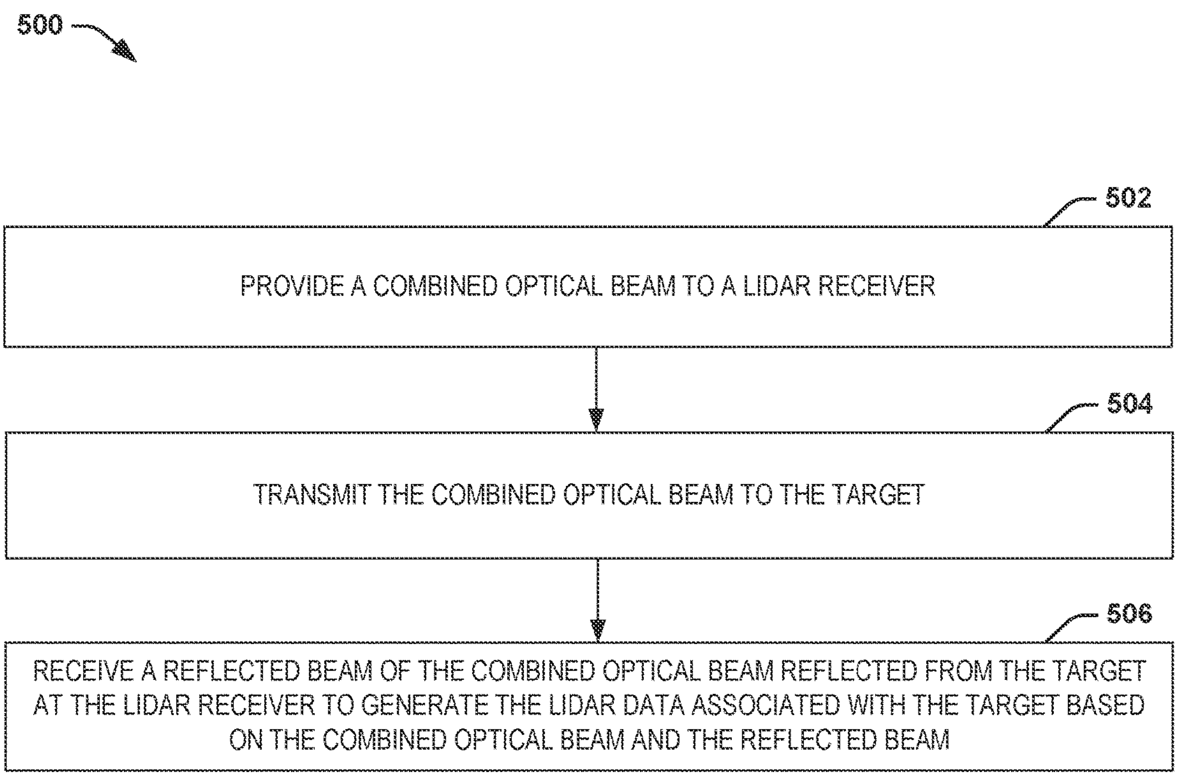
500
502
PROVIDE A COMBINED OPTICAL BEAM TO A LIDAR RECEIVER
504
TRANSMIT THE COMBINED OPTICAL BEAM TO THE TARGET
506
RECEIVE A REFLECTED BEAM OF THE COMBINED OPTICAL BEAM REFLECTED FROM THE TARGET AT THE LIDAR RECEIVER TO GENERATE THE LIDAR DATA ASSOCIATED WITH THE TARGET BASED ON THE COMBINED OPTICAL BEAM AND THE REFLECTED BEAM

FIG. 5

QUANTUM LIDAR SYSTEM

TECHNICAL FIELD

The present invention relates generally to lidar systems, and specifically to a quantum lidar system.

BACKGROUND

Lidar is a type of sensor that can provide range-finding and/or imaging based on a laser. As an example, a lidar system can determining ranges (variable distance) by targeting an object with a laser and measuring the time for the reflected light to return to the receiver. A lidar system can also be used to create digital three-dimensional images of areas on terrestrial surfaces, on the ocean floor, and/or structures (e.g., buildings) thereon due to differences in laser return times and by varying laser wavelengths. A quantum lidar system can typically implement a nonlinear device to degeneratively create a signal beam and an idler beam from a single optical pump beam, such that the signal beam provided to the target and the idler beam provided to a local reference can be used to generate lidar data associated with the target.

SUMMARY

One example includes a quantum lidar system. The system includes a beam generator configured to generate a signal beam and an idler beam and a beam combiner configured to generate a combined optical beam comprising the signal beam and the idler beam. The system also includes a lidar transmitter configured to transmit the combined optical beam to a target and a lidar receiver configured to receive the combined optical beam and a reflected beam of the combined optical beam reflected from the target to generate lidar data associated with the target.

Another example includes a method for generating lidar data associated with a target. The method includes providing the combined optical beam to a lidar receiver. The method also includes transmitting the combined optical beam to the target, and receiving a reflected beam of the combined optical beam reflected from the target at the lidar receiver to generate the lidar data associated with the target based on the combined optical beam and the reflected beam.

Another example includes a quantum lidar system. The system includes a beam generator configured to generate a signal beam and an idler beam and a beam combiner configured to generate a combined optical beam comprising the signal beam and the idler beam. The signal beam and the idler beam can have unequal frequencies. The system also includes a lidar transmitter configured to transmit the combined optical beam to a target. The system further includes a lidar receiver. The lidar receiver includes a local detector configured to receive the combined optical beam and to generate a first detection signal associated with the combined optical beam. The lidar receiver also includes a target detector configured to receive a reflected beam of the combined optical beam reflected from the target and to generate a second detection signal associated with the reflected beam. The lidar receiver further includes a lidar processor configured to generate lidar data associated with the target based on the first and second detection signals.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** illustrates an example block diagram of a quantum lidar system.

FIG. **2** illustrates another example block diagram of a quantum lidar system.

FIG. **3** illustrates an example block diagram of a beam generator and beam combiner.

FIG. **4** illustrates an example block diagram of a lidar receiver.

FIG. **5** illustrates an example of a method for generating lidar data associated with a target.

DETAILED DESCRIPTION

The present invention relates generally to lidar systems, and specifically to a quantum lidar system. The quantum lidar system can be implemented for any of a variety of applications for range-finding and/or imaging. The quantum lidar system includes a beam generator that is configured to generate a signal beam and an idler beam. As an example, the signal and idler beams can be degeneratively created based on providing an optical pump beam through a nonlinear device. As an example, the nonlinear device can be a spontaneous parametric downconverter (SPDC) device, such as to generate the signal beam and the idler beams as having different frequencies. The quantum lidar system also includes a beam combiner that is configured to combine the signal beam and the idler beam to generate a combined optical beam. The beam combiner can be configured as a set of optics that can generate the combined optical beam and provide the combined optical beam to a lidar transmitter and to a lidar receiver.

The lidar transmitter can be configured to provide the combined optical beam to a target. The combined optical beam can be reflected from the target to provide a reflected beam. The lidar receiver can receive both the combined optical beam (e.g., from the beam combiner) and the reflected beam and can generate lidar data associated with the target based on the combined optical beam and the reflected beam. For example, the lidar receiver can include a lidar processor that can implement a delayed choice temporal convolution algorithm to generate the lidar data. Based on the quantum entanglement of the signal and idler beams in the combined optical beam provided to both the target and the lidar receiver, the lidar processor can greatly increase a signal-to-noise ratio (SNR) of the resultant lidar data based on determining a convolution peak of one of the signal and idler beams in the combined optical signal provided to the lidar receiver and the other of the signal and idler beams in the reflected beam provided to the lidar receiver.

FIG. **1** illustrates an example block diagram of a quantum lidar system **100**. The quantum lidar system **100** can be implemented in any of a variety of range-finding and/or imaging applications with respect to a target **102**. The target **102** can correspond to geographic features (e.g., terrestrial, underwater, and/or surfaces of other celestial bodies) or to man-made structures, such as buildings or vehicles. Thus, the quantum lidar system **100** can be configured to determine a range to the target **102** and/or generate image data associated with the target **102**.

The quantum lidar system **100** includes a beam generator **104** and a beam combiner **106**. The beam generator **104** is configured to generate a signal beam and an idler beam that can be implemented in a quantum entangled beam (e.g., via a nonlinear device). The beam combiner **106** is thus configured to combine the signal beam and the idler beam to generate the quantum entangled beam, described hereinafter as a "combined optical beam". The combined optical beam, demonstrated in the example of FIG. **1** as a beam

OPTCMBD, can therefore include both the signal beam and the idler beam having a common wavefront. As an example, the beam combiner **106** can include a variety of different types of optical devices to combine the signal beam and the idler beam. As described herein, the beam combiner **106** can generate the combined optical beam OPT_{CMBD} such that the signal beam and the idler beam have a single photon entanglement.

The beam combiner **106** can provide the combined optical beam OPT_{CMBD} to a lidar transmitter **108** and a lidar receiver **110**. The lidar transmitter **108** is therefore configured to illuminate the target **102** with the combined optical beam OPT_{CMBD}. The combined optical beam OPT_{CMBD} is thus reflected from the target **102** and provided back to quantum lidar system **100** as a reflected beam OPT_{RFL} to be received by the lidar receiver **110**. The lidar receiver **110** thus receives both the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL}, such that the lidar receiver **110** is configured to generate lidar data associated with the target **102** based on the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL}. As an example, the lidar receiver **110** can include a lidar processor that is configured to implement a temporal convolution algorithm on the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL} to generate the lidar data associated with the target **102**. The temporal convolution algorithm can be a delayed choice detection algorithm, such that the temporal convolution algorithm can be implemented at a time after receipt of the reflected beam OPT_{RFL}.

As an example, the lidar receiver **110** can include a local detector that is configured to monitor the combined optical beam OPT_{CMBD} and a target detector that is configured to monitor the reflected beam OPT_{RFL}. Because the combined optical beam OPT_{CMBD} includes both the signal beam and the idler beam, the reflected beam OPT_{RFL} likewise includes the signal beam and the idler beam reflected back from the target **102**. Therefore, the local detector of the lidar receiver **110** can agnostically detect one of the signal beam and the idler beam in the combined optical beam OPT_{CMBD}, and the target detector of the lidar receiver can agnostically detect the other one of the signal beam and the idler beam in the reflected beam OPT_{RFL}. As a result, the lidar processor can implement the temporal convolution algorithm in a manner that increases signal-to-noise ratio (SNR) for a stronger correlation between the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL}, and which obviates the need for a quantum memory. As a result, the quantum lidar system **100** can have a simpler design without the quantum memory, and can provide for more accurate lidar data associated with the target **102**.

FIG. **2** illustrates another example block diagram of a quantum lidar system **200**. The quantum lidar system **200** can correspond to the quantum lidar system **100** in the example of FIG. **1**. Therefore, reference is to be made to the example of FIG. **1** in the following description of the example of FIG. **2**.

The quantum lidar system **200** includes a beam generator **202**. The beam generator **202** is configured to generate a signal beam OPT_{SGNL} and an idler beam OPT_{IDLR} that can be implemented in a quantum entangled beam. As an example, the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} can be generated based on a parametric degeneration process from an optical pump beam provided through a nonlinear device. For example, the beam generator **202** can include a spontaneous parametric downconverter (SPDC) to generate the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} from the optical pump beam. Therefore, the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} can be generated to have unequal frequencies. Therefore, the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} can be separately detected, as described in greater detail herein.

The quantum lidar system **200** also includes a beam combiner **204** that is configured to combine the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} to generate the combined optical beam OPT_{CMBD}. The combined optical beam OPT_{CMBD} can therefore include both the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} having a common wavefront. As an example, the beam combiner **204** can include a variety of different types of optical devices to combine the signal beam and the idler beam. As another example, the beam combiner **204** can be combined with the beam generator **202**, such that the beam generator **202** and the beam combiner **204** can include a set of optics that operate together to generate the combined optical beam OPT_{CMBD} that includes the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR}. As an example, the beam combiner **204** can generate the combined optical beam OPT_{CMBD} such that the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} have a single photon entanglement.

The beam combiner **204** can provide the combined optical beam OPT_{CMBD} to a lidar transmitter **206** and a lidar receiver **208** (e.g., via optics). The lidar transmitter **206** is therefore configured to illuminate a target (e.g., the target **102**) with the combined optical beam OPT_{CMBD}. The combined optical beam OPT_{CMBD} is thus reflected from the target and provided back to quantum lidar system **200** as a reflected beam OPT_{RFL} to be received by the lidar receiver **208**. The lidar receiver **208** thus receives both the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL}, such that the lidar receiver **208** is configured to generate lidar data associated with the target based on the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL}. As an example, the lidar receiver **208** can include a lidar processor that is configured to implement a temporal convolution algorithm on the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL} to generate the lidar data associated with the target. The temporal convolution algorithm can be a delayed choice detection algorithm, such that the temporal convolution algorithm can be implemented at a time after receipt of the reflected beam OPT_{RFL}.

FIG. **3** illustrates an example block diagram of a beam generator **300**. The beam generator **300** can correspond to the beam generators **104** and **202** of the respective examples of FIGS. **1** and **2**. Therefore reference is to be made to the examples of FIGS. **1** and **2** in the following description of the example of FIG. **3**.

The beam generator **300** includes a beam source **302** that is configured to generate an optical pump beam OPT_{PMP}. The beam source **302** can be a laser, waveguide, optical fiber, or other component to provide the optical pump beam OPT_{PMP}. The beam generator **300** also includes an SPDC **304** that is configured to degeneratively create the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} from the optical pump beam OPT_{PMP}. For example, the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} can have unequal frequencies. While the example of FIG. **3** demonstrates that the nonlinear device that generates the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} is the SPDC **304**, other types of nonlinear devices could be used instead. Additionally, as one example, the optical pump beam OPT_{PMP} can be provided in a single pass through the SPDC **304** to provide for a single photon entanglement of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR}.

FIG. 4 illustrates an example block diagram of a lidar receiver 400. The lidar receiver 400 can correspond to the lidar receiver 110 and the lidar receiver 208 in the respective examples of FIGS. 1 and 2. Therefore, reference is to be made to the examples of FIGS. 1 and 2 in the following description of the example of FIG. 4.

The lidar receiver 400 includes a local detector 402. The local detector 402 is configured to receive the combined optical beam OPT_{CMBD} from the beam combiner (e.g., via optics). The local detector 402 includes a local photon counter 404 that is configured to monitor photons associated with either the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} of the combined optical beam OPT_{CMBD}. As an example, the local photon counter 404 can be tuned to the frequencies of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR}, such that the local photon counter 404 can agnostically identify one of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} in the combined optical beam OPT_{CMBD}.

The lidar receiver 400 also includes a target detector 406. The target detector 406 is configured to receive the reflected beam OPT_{RFL} reflected back from the target (e.g., the target 102). The target detector 406 includes a target photon counter 408 that is configured to monitor photons associated with either the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} portions of the reflected beam OPT_{RFL}. As an example, the target photon counter 408 can be tuned to the frequencies of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR}, such that the target photon counter 408 can agnostically identify one of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} in the reflected beam OPT_{RFL}. The one of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} that is detected by the target detector 406 is the opposite of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} that is detected by the local detector 402. Therefore, the lidar receiver 400 can find a correlation between one of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} in one of the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL} and the other of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} in the other of the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL}. For example, the quantum lidar system can exhibit a range accuracy of $0.5*C*\Delta T$, where C is the speed of light and ΔT is the response time of the local and target detectors 402 and 406.

In the example of FIG. 4, the local detector 402 is configured to generate a first detection signal DET_1 that corresponds to the respective one of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} in the combined optical beam OPT_{CMBD}. Similarly, the target detector 406 is configured to generate a second detection signal DET_2 that corresponds to the respective other one of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} in the reflected beam OPT_{RFL}. The detection signals DET_1 and DET_2 are provided to a lidar processor 410 that is configured to generate the lidar data associated with the target based on the detection signals DET_1 and DET_2.

As an example, the lidar processor 410 can implement a delayed choice temporal convolution algorithm on the detection signals DET_1 and DET_2 to generate the lidar data. For example, the lidar processor 410 can provide temporal synchronization of the local detector 402 and the target detector 406 over a period of time and store the time bins corresponding to the detection signals DET_1 and DET_2 to implement a delayed choice processing of the detection signals DET_1 and DET_2. The lidar processor 410 can then convolve the detection signals DET_1 and DET_2 with a variable time delay, such that sweeping the time delay across the range gate can provide a convolution peak corresponding to the time bins of maximum correlation between the detection signals DET_1 and DET_2. The time offset that corresponds to the convolution peak can thus correspond to the range to the target.

As described herein, the combined optical beam OPT_{CMBD} that is provided to the local detector 402 includes both the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR}, and the combined optical beam OPT_{CMBD} that is reflected from the target back to the target detector 406 likewise includes both the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR}. Based on implementing the quantized representation of the electric fields for the photons of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR}, and by calculating the correlation between the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL}, only the entangled photons contribute to the convolution. Thus, only one of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} of the combined optical beam OPT_{CMBD} and the other one of the signal beam OPT_{SGNL} and the idler beam OPT_{IDLR} of the reflected beam OPT_{RFL} contribute to the convolution, as determined at the local detector 402 and the target detector 406. Accordingly, the lidar processor 410 can determine the lidar data at a significantly higher SNR relative to typical lidar processing algorithms. Furthermore, because the time synchronized detections of the combined optical beam OPT_{CMBD} and the reflected beam OPT_{RFL} are performed separately in the delayed choice manner, the lidar receiver 400 does not need a quantum memory to determine the lidar data associated with the target.

In view of the foregoing structural and functional features described above, a methodology in accordance with various aspects of the disclosure will be better appreciated with reference to FIG. 5. It is to be understood and appreciated that the method of FIG. 5 is not limited by the illustrated order, as some aspects could, in accordance with the present disclosure, occur in different orders and/or concurrently with other aspects from that shown and described herein. Moreover, not all illustrated features may be required to implement a methodology in accordance with an aspect of the present examples.

FIG. 5 illustrates an example of a method 500 for generating lidar data associated with a target (e.g., the target 102). At 502, a combined optical beam (e.g., the combined optical beam OPT_{CMBD}) is provided to a lidar receiver (e.g., the lidar receiver 110). The combined optical beam includes a signal beam and an idler beam. At 504, the combined optical beam is transmitted to the target. At 506, a reflected beam (e.g., the reflected beam OPT_{RFL}) of the combined optical beam reflected from the target is received at the lidar receiver to generate the lidar data associated with the target based on the combined optical beam and the reflected beam.

What have been described above are examples of the present invention. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the present invention, but one of ordinary skill in the art will recognize that many further combinations and permutations of the present invention are possible. Accordingly, the present invention is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Additionally, where the disclosure or claims recite "a," "an," "a first," or "another" element, or the equivalent thereof, it should be interpreted to include one or more than one such element, neither requiring nor excluding two or more such elements. As used herein, the term "includes" means includes but not limited to, and the term "including" means including but not limited to. The term "based on" means based at least in part on.

What is claimed is:

1. A quantum lidar system comprising:
a beam generator configured to generate a signal beam and an idler beam;
a beam combiner configured to generate a combined optical beam comprising the signal beam and the idler beam;
a lidar transmitter configured to transmit the combined optical beam to a target; and
a lidar receiver configured to receive the combined optical beam and a reflected beam of the combined optical beam reflected from the target to generate lidar data associated with the target.

2. The system of claim **1**, wherein the beam generator comprises:
a beam source configured to generate a pump beam; and
a nonlinear device configured to degenerate the pump beam into the signal beam and the idler beam.

3. The system of claim **2**, wherein the nonlinear device is arranged as a spontaneous parametric down-conversion device.

4. The system of claim **2**, wherein the nonlinear device is configured to degenerate the pump beam into the signal beam having a first frequency and the idler beam having a second frequency that is not equal to the first frequency.

5. The system of claim **1**, wherein the beam combiner comprises a set of optics configured to combine the signal beam and the idler beam to generate the combined optical beam, and to provide the combined optical beam to the lidar transmitter and the lidar receiver.

6. The system of claim **1**, wherein the lidar receiver comprises:
a local detector configured to receive the combined optical beam and to generate a first detection signal associated with the combined optical beam;
a target detector configured to receive the reflected beam and to generate a second detection signal associated with the reflected beam; and
a lidar processor configured to generate the lidar data based on the first and second detection signals.

7. The system of claim **6**, wherein the lidar processor is configured to implement a temporal convolution algorithm on the first detection signal and the second detection signal to generate the lidar data.

8. The system of claim **7**, wherein the lidar processor is configured to implement the temporal convolution algorithm as a delayed choice detection algorithm.

9. The system of claim **7**, wherein the temporal convolution algorithm is configured to determine a convolution peak between one of the signal beam and the idler beam associated with the combined optical beam and the other of the signal beam and the idler beam associated with the reflected optical beam.

10. A method for generating lidar data associated with a target, the method comprising:
providing a combined optical beam to a lidar receiver, the combined optical beam comprising a signal beam and an idler beam;
transmitting the combined optical beam to the target; and
receiving a reflected beam of the combined optical beam reflected from the target at the lidar receiver to generate the lidar data associated with the target based on the combined optical beam and the reflected beam.

11. The method of claim **10**, further comprising:
generating a pump beam; and
degenerating the pump beam into the signal beam and the idler beam via a nonlinear device.

12. The method of claim **11**, wherein degenerating the pump beam comprises degenerating the pump beam into the signal beam and the idler beam via a spontaneous parametric down-conversion device.

13. The method of claim **11**, wherein degenerating the pump beam comprises degenerating the pump beam into the signal beam having a first frequency and the idler beam having a second frequency that is not equal to the first frequency.

14. The method of claim **11**, further comprising:
generating a first detection signal associated with the combined optical beam provided to the lidar receiver;
generating a second detection signal in response to receiving the reflected beam; and
implementing a temporal convolution algorithm on the first detection signal and the second detection signal to generate the lidar data.

15. The method of claim **14**, wherein implementing the temporal convolution algorithm comprises implementing the temporal convolution algorithm as a delayed choice detection algorithm.

16. A quantum lidar system comprising:
a beam generator configured to generate a signal beam and an idler beam;
a beam combiner configured to generate a combined optical beam comprising the signal beam and the idler beam, wherein the signal beam and the idler beam have unequal frequencies;
a lidar transmitter configured to transmit the combined optical beam to a target; and
a lidar receiver comprising:
a local detector configured to receive the combined optical beam and to generate a first detection signal associated with the combined optical beam;
a target detector configured to receive a reflected beam of the combined optical beam reflected from the target and to generate a second detection signal associated with the reflected beam; and
a lidar processor configured to generate lidar data associated with the target based on the first and second detection signals.

17. The system of claim **16**, wherein the beam generator comprises:
a beam source configured to generate a pump beam; and
a spontaneous parametric down-conversion device configured to degenerate the pump beam into the signal beam having a first frequency and the idler beam having a second frequency that is not equal to the first frequency.

18. The system of claim **16**, wherein the lidar processor is configured to implement a temporal convolution algorithm on the first detection signal and the second detection signal to generate the lidar data.

19. The system of claim **18**, wherein the lidar processor is configured to implement the temporal convolution algorithm as a delayed choice detection algorithm.

20. The system of claim **18**, wherein the temporal convolution algorithm is configured to determine a convolution peak between one of the signal beam and the idler beam associated with the combined optical beam and the other of the signal beam and the idler beam associated with the reflected optical beam.

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